

10 Questions — Method Overriding

Q1 — Easy

Create class Animal with method speak()
Override in Dog, Cat, Cow, Sheep
Each animal speaks differently
Loop all objects and call speak()

Q2 — Easy

Create class Shape with method draw()
Override in Circle, Rectangle, Triangle, Square
Each shape draws differently with emoji
Loop all shapes and call draw()

Q3 — Easy

Create class Phone with methods call() and message()
Override in Samsung, iPhone, OnePlus, Nokia
Each phone has its own calling and messaging style
Create one object each and call both methods

Q4 — Medium

Create class Vehicle with methods fuel_type() and move()
Override in Car, Bike, Boat, Airplane, Rocket
Each has different fuel and movement
Add method show() in parent that calls both
Loop all and call show()

Q5 — Medium

Create class Employee with methods work() and calculate_bonus()

- work() → prints working message

- calculate_bonus() → returns 5% of salary
Override both in Developer, Designer, Manager, Intern
Each has different work message and bonus percentage
Show who gets highest bonus at end
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Q6 — Medium

Create class Food with methods prepare() and serve()
Override in Pizza, Burger, Sushi, Pasta
Add method order() in parent that calls both
Each food has different preparation and serving style
Create **2 objects** of each food and call order()

Q7 — Medium

Create class Bank with methods:

- account_type() → prints "Basic Account"
 - interest_rate() → returns 0%
 - show() → prints owner, balance, type, interest
Override in SavingsAccount, CurrentAccount,
FixedDeposit, LoanAccount
Each has different type and interest rate
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Q8 — Hard

Create class Person with private __name, __age
Add methods introduce() and show_role()
Override both in Student, Teacher, Doctor, Engineer
Each has own private attributes added on top
Add validation in mutators
Each introduces differently and has different role message

Q9 — Hard

Create class Game with methods:

- start() → prints "Game Starting..."
 - play() → prints "Playing game"
 - score() → returns 0
 - end() → prints "Game Over"
- Override all methods in:
Cricket, Football, Basketball, Tennis
Each game starts, plays, scores, and ends differently
Add class attribute total_games = 0
Add class method get_total() → increases on each object
Show total games at end
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Q10 — All Together

Create class Vehicle with:

- private __brand, __speed, __price
- accessors + mutators with validation
- methods fuel_type(), move(), maintenance()
- static method is_valid_speed(speed) → 0 to 500
- class attribute total_vehicles = 0
- class method get_total()
- method show() → prints all details + calls all 3 methods

Override fuel_type(), move(), maintenance() in:

Car, ElectricCar, Motorcycle, Truck, Airplane

Each has:

- own private attribute added on top
- own version of all 3 overridden methods
- own maintenance cost and style

Create one object of each

Test all validations

Show total vehicles at end

